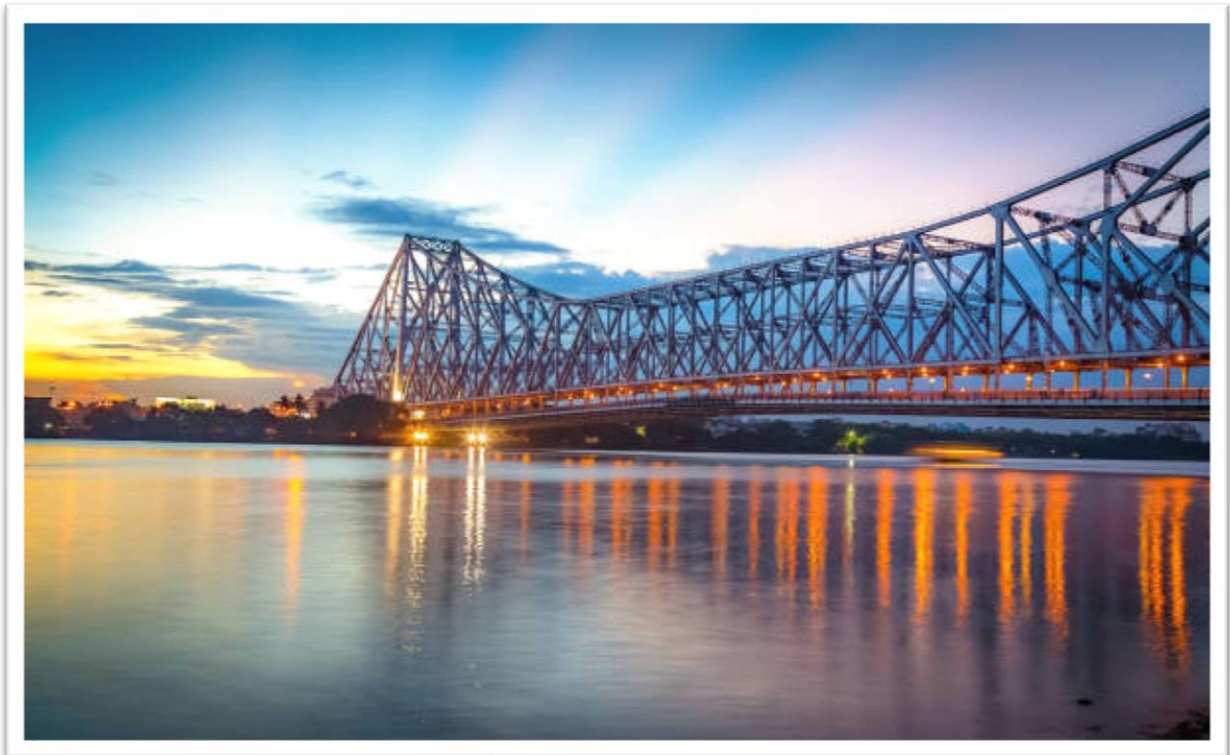


Trusst-It!

(Let's Bring out the Structural Engineer in us and Build a Simple Truss Bridge)



I. Some Funda Behind Bridges

You might have seen these bridges at many places including the famous Howrah bridge



Wondered the engineering behind it? It's quite interesting & easy to grasp the basics. Watch these two videos and learn the basic engineering detail that civil & structural engineers use to build these bridges.

1. [Why do bridges have triangular shapes?](#)
2. [Why are Triangles Stronger than Squares?](#)

II: Let's Build a Virtual Truss Bridge (Simulation)

Open: John Hopkins Truss Simulator linked here

Create a Virtual Truss Bridge with [this Reference Video](#) and make a note of the members that are subjected to high tensile and compressive stress when load (Vertical force F) on the bridge is increased.

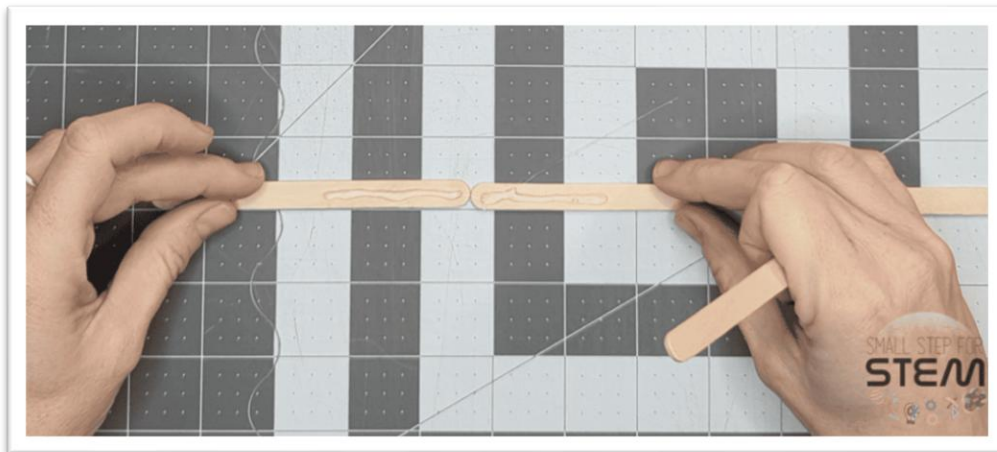
Now let's test the simulation with real testing.

III. Let's Build a Mini version Truss Bridge (Physical)

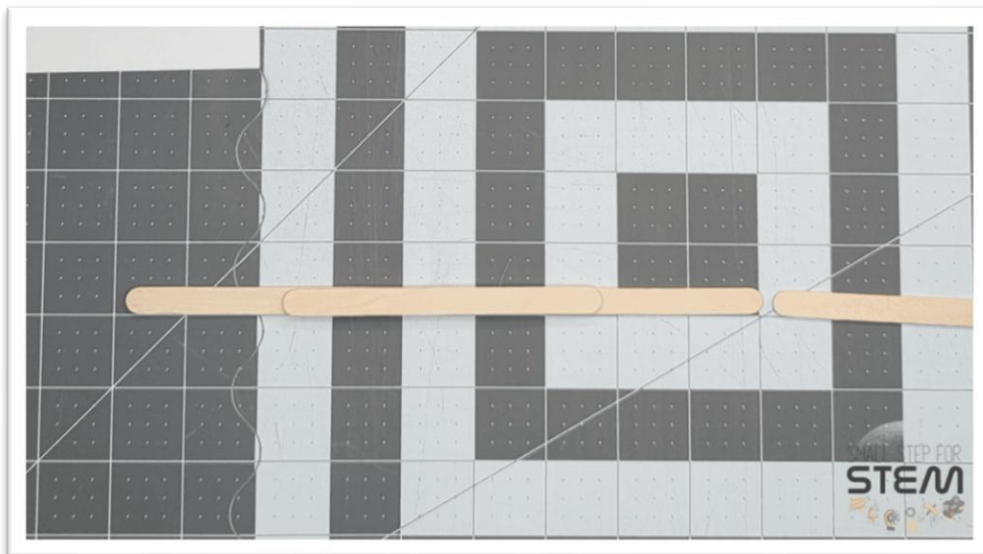
Estimated time : 30-45 mins as team activity

(Courtesy : [Small steps for STEM](#)) .

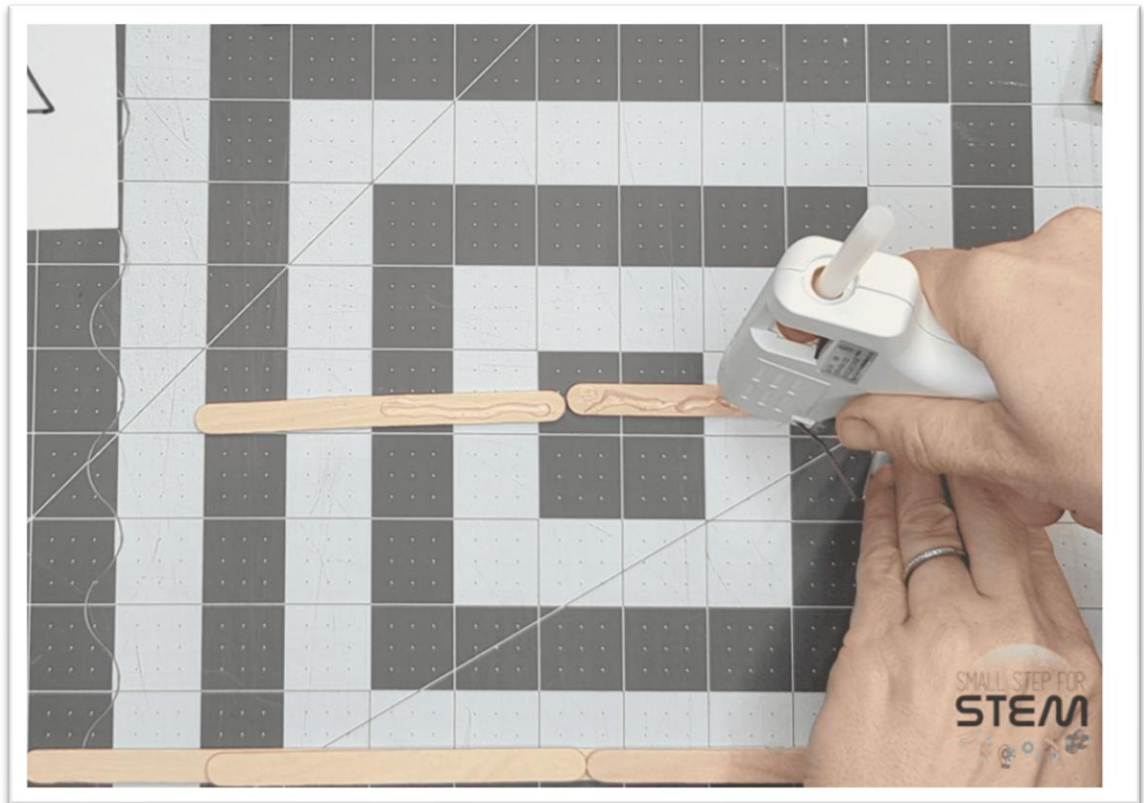
Step 1: Make the Top & Bottom Beams of the bridge as shown below



Glue applied at the edges as shown

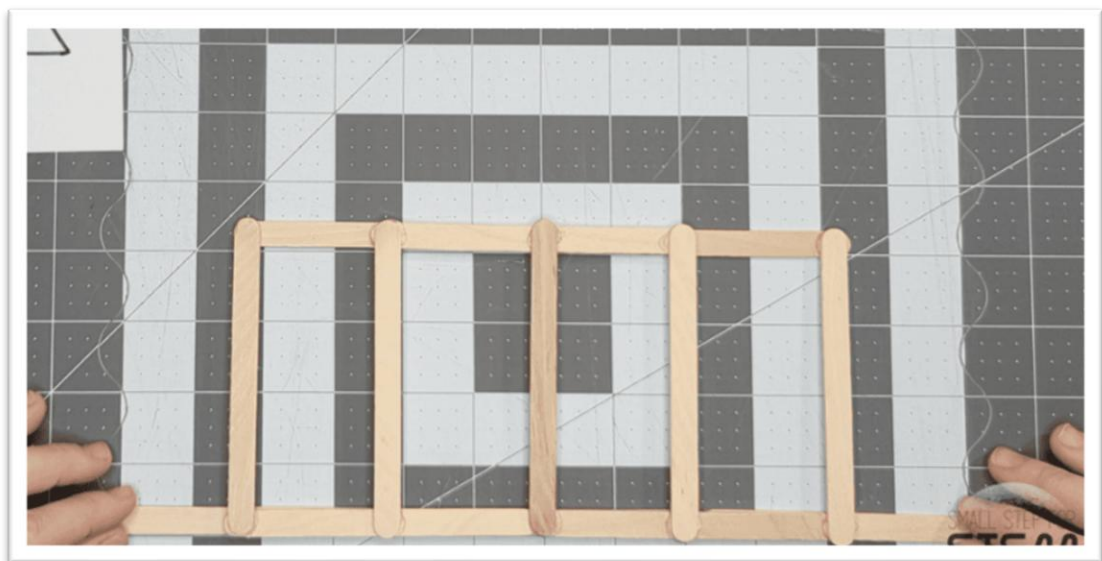


Both Sticks are Joined and make another joint with one more stick
(3 base sticks with 2 connecting sticks)



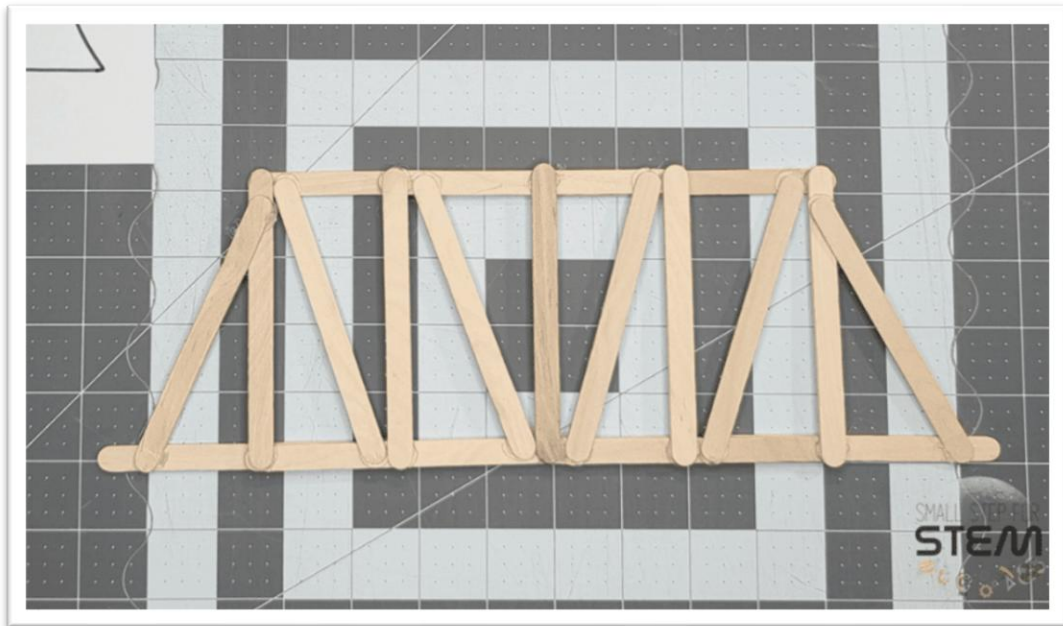
Follow the same approach for Top Beam (with one less stick)
(2 base stick with one connecting stick)

Step 2 : Connect the Top and Bottom Beam with Vertical Members



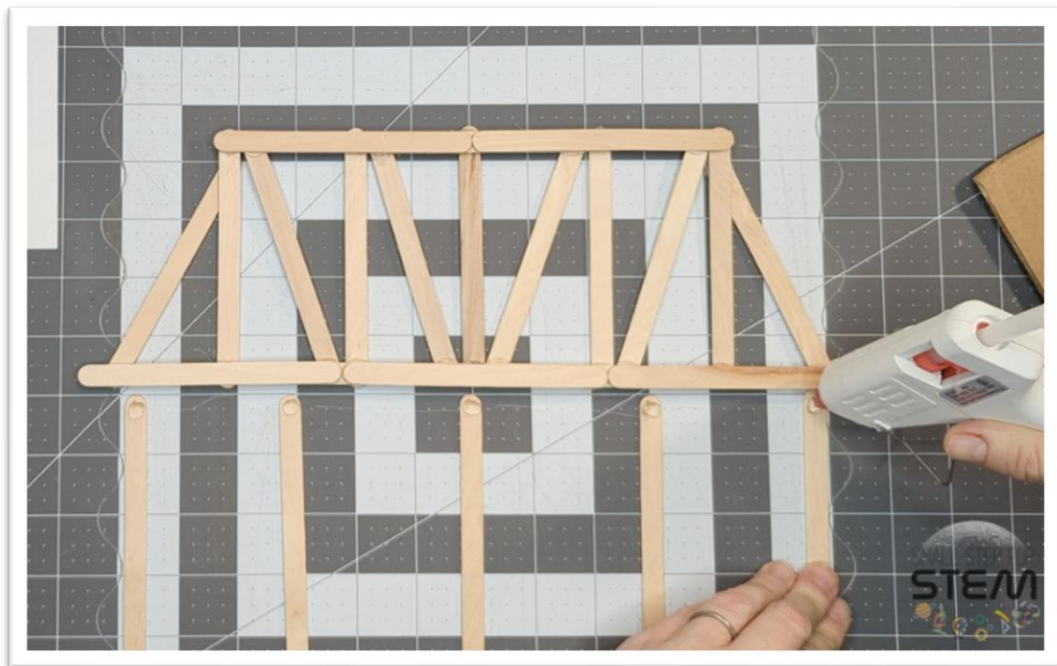
(Total 5 Vertical Members as shown)

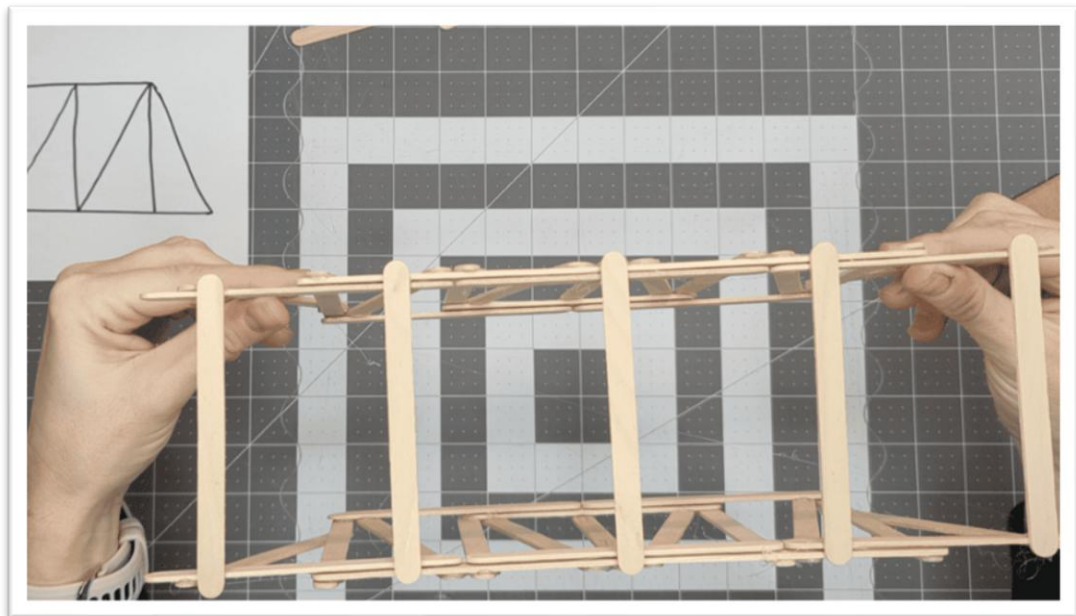
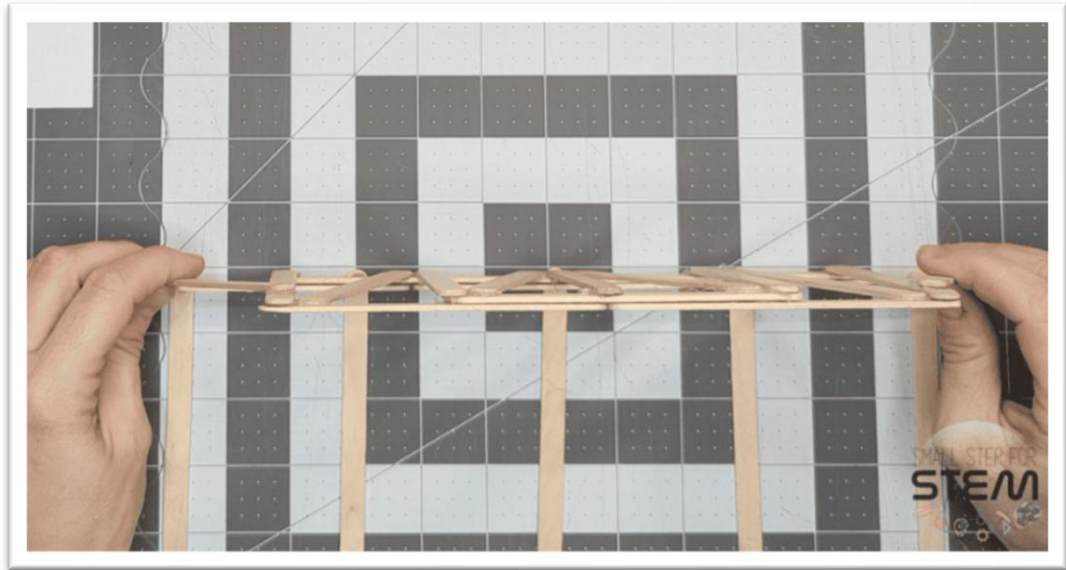
Step 3 : Make the Angle Members as shown

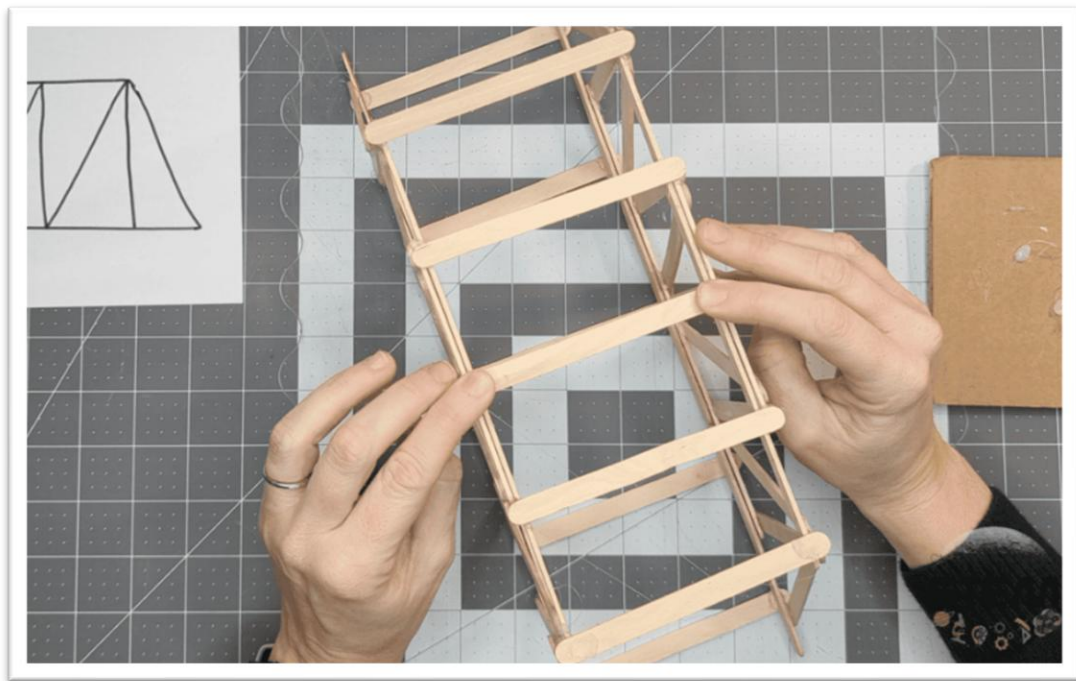


Step 4: Repeat Steps 1-3 and make the other side of the bridge

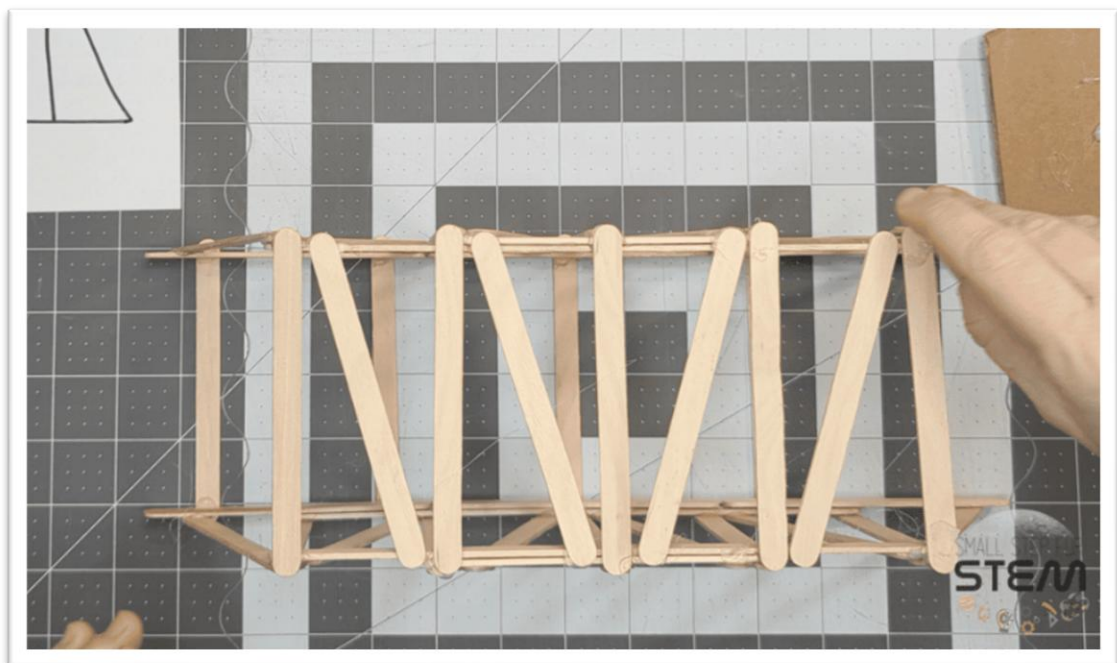
Step 5: Connect the two sides as shown







Step 6 : Add Angled Support for top side as shown



III. Time for Testing

Gather your friends, load the bridge from top slowly, see the fun unload!

Importantly make a note of the member that fails (breaks off) and cross check with the simulated member with high stress.

